



Overview Paper

Evacuation transportation modeling: An overview of research, development, and practice

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ABSTRACT

This paper presents a review of highway-based evacuation modeling and simulation and its evolution over the past decade. The review includes the major components of roadway transportation planning and operations, including the current state of modeling in the forecasting of evacuation travel demand, distribution and assignment of evacuation demand to regional road networks to reach destinations, assignment of evacuees to various modes of transportation, and evaluation and testing of alternative management strategies to increase capacity of evacuation networks or manage demand. Although this discussion does not cover recent work in other modes used in evacuation such as air, rail, and pedestrian, this paper does highlight recent interdisciplinary modeling work in evacuation to help bridge the gap between the behavioral sciences and engineering and the application of emerging techniques for the verification, validation, and calibration of models. The manuscript also calls attention to special considerations and logistical difficulties, which have received limited attention to date. In addition to these concerns, the following future directions are discussed: further interdisciplinary efforts, including incorporating the medical community; using new technologies for communication of warnings and traffic condition information, data collection, and increased modeling resolution and confidence; using real-time information; and further model refinements and validation.

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1. Introduction

Evacuations, while often thought of as rare events, are in reality more common. A study conducted by the Sandia National Laboratory for the United States (US) Nuclear Regulatory Commission (NRC) showed that, on average, an evacuation of 1000 or more people occurs about once every two to three weeks in the US (Sandia National Laboratories, 2004). Also notable in the NRC data is the high frequency (nearly 75% of the total sample) of small-scale evacuations involving less than 5000 people. These data, along with the range of hazards for which evacuations are ordered, suggest that evacuations can be needed virtually anywhere and must often be carried out with little advanced warning time to mobilize and implement traffic control and management measures. The realization of these trends led to an increased level of involvement from transportation professionals in emergency planning and response over the past 10–15 years. Today, the roles and responsibilities of transportation during emergencies have been formalized in the National Response Framework (Department of Homeland Security, 2008).

Among transportation's numerous roles is the planning of transportation assets and infrastructure to respond to major emergencies, including the management of evacuation and detour routes; utilization of mass transit systems; movement

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and distribution of relief supplies; and the restoration of damaged transportation systems. Transportation engineers have also been involved in the development and application of modeling and simulation systems to support the development of evacuation plans. The past several decades have seen significant advances in both the number and sophistication of systems adapted for and dedicated specifically to the purpose of evacuation modeling and simulation. Several of the earliest efforts evolved from the analysis of nuclear power plant emergencies in the aftermath of the Three Mile Island emergency then continued to evolve in detail and complexity throughout the next two decades, particularly for hurricane evacuations. Most recently, and in parallel with enormous advances in the computational speed and power of computers, the history altering events of September 11, 2001 and Hurricanes Katrina and Rita in 2005 have led to a greatly increased level of interest and involvement in evacuation modeling from within the transportation research community.

This paper presents a review of highway-based evacuation modeling and simulation and its evolution over the past decade. The review includes the major components of areas of roadway transportation planning and operations, including the current state of modeling in the:

- forecasting of evacuation travel demand;
- the distribution and assignment of evacuation demand to regional road networks to reach destinations;
- assignment of evacuees to various modes of transportation; and
- evaluation and testing of alternative management strategies to assess and increase capacity of evacuation networks.

Although this discussion does not cover recent work in other modes use in evacuation such as air, rail, and pedestrian (each of which has garnered increasing levels of attention and have seen significant advancements of their own), this paper does highlight recent interdisciplinary modeling work in evacuation to help bridge the gap between the behavioral sciences and engineering and the application of emerging techniques for the verification, validation, and calibration of models.

The discussion is organized into seven additional sections. The first provides an overview of evacuation warnings and information dissemination. Section 3 discusses zoning for evacuation warning dissemination and demand modeling. Section 4 presents a wide perspective on demand modeling, including social and threat factors influencing the decision to evacuate, trip generation techniques, departure time modeling, and mode split/choice. Section 5 provides an overview of route selection and traffic assignment. Section 6 discusses a selection of strategies to make evacuations more efficient. Section 7 describes some special logistical considerations that further complicate the evacuation process. Finally, Section 8 presents future needs and directions.

2. Warning and information

Among the most critical needs in carrying out an evacuation is the clear and effective communication of evacuation orders. This communication must overcome the disbelief that is typically the initial response to disaster warnings (Drabek, 1999; Tierney et al., 2001). The wording and content of evacuation orders (message), person delivering the message (source), and distribution medium (channel) (Lindell and Perry, 2004, 2012), can heavily influence not only the number of people that evacuate, but also the urgency at which they leave, the areas from which they depart, and the destinations that they chose. Drabek (1999) suggests that effective messages answer the questions “(1) Who is issuing the warning? (2) What is threatening? (3) What exact geographical area is threatened? (4) When is it coming? (5) How probable is the event? (6) Are there high risk locations, such as people in automobiles, that require special actions? (7) What specific protective actions should be taken?” (p. 520).

Interestingly, there is little consistency in terminology from location to location and, often, wide variation within a single location. Acronyms and jargon used in social media and the Internet may further cause consistency issues. A recent study of evacuation practices showed the range of wording used by public officials when issuing evacuation orders (Wolshon, 2009) as well as the intended meaning they are meant to convey. Among survey respondents, the terms “Mandatory” and “Voluntary” evacuation were the most common. It is notable, however, that the definition of “mandatory” is legally unclear (Baker, 1991) because emergency management and law enforcement agencies realize that it is not realistic to enforce an order that compels people to evacuate. However, the term “mandatory” carries significant weight with potential evacuees in recommending that they evacuate.

Disaster warnings are considered a social process, involving the interaction of warning sources and message receivers and receivers with each other and others in their social networks (e.g., family). For example, the messages can reach the ultimate receivers either directly from the source or through intermediate sources via a diffusion process. The information must be received and understood before action is likely to occur (Lindell and Perry, 2012), particularly in the absence of environmental or social cues. People interpret messages differently and vary in their beliefs of the messages and subsequent actions (Drabek, 1986). Personal characteristics and those of the channel, source, and message have a complex relationship that affects how the information is perceived and understood. People often seek confirmation of the information from other sources (e.g., friends, relatives, neighbors, coworkers (Tierney et al., 2001)) or additional warning messages (Drabek, 1986). In ambiguous situations, people may be guided by reference groups and even strangers (Lindell and Perry, 2004). This confirmation helps overcome initial inertia or disbelief. Perry (1985) presents a sequential flow chart of the decision making process for responding to natural disaster warnings that involves (1) receiving an initial message, (2) interpreting the message, (3) assessing personal risk, (4) determining whether protection is feasible, (5) determining whether protective action can be

undertaken, (6) determining whether the action will significantly reduce the consequences, (7) assessing alternatives, and (8) selecting an action. If protective actions are not feasible or insignificantly affect consequences, no action is chosen. Later decision making process representations include the Protective Action Decision Model (Lindell and Perry, 1992, 2012), which involves a feedback loop (Lindell and Perry, 2012), and Mileti and Sorensen's (1990) model.

Once evacuees take to the road, the conveyance of timely, accurate, and useful information to evacuees is vital to carrying out the evacuation process. Recent research conducted in South Carolina examined the role of communication during hurricane evacuations as well as the communication modes that are most effective to convey information before and during the evacuation (Taaffe et al., 2012 (under editorial review)). The results of the study suggested that the type and methods of communication currently used in South Carolina to order and convey evacuation information results in a more vigilant and prepared population than was apparent during Hurricane Hugo in 1989. Not surprisingly, the pervasive use of cell phones and the Internet has been one of the most significant improvements, particularly for the rapid and wide distribution of emergency information through online and other forms of digital media.

The role of social media in sharing information about disasters is gaining particular attention. Sutton et al. (2008), for example, used an online questionnaire about information and communications technology use and information gathering and sharing activities to capture perishable data. They found that the majority of their respondents sought information online and 36% posted information or were part of online discussion groups. Some of this information seeking was to fill gaps in official news sources or obtain more local information.

The use of new information distribution technologies for emergency notification and information sharing will continue to evolve with the technologies themselves. However, one should remain aware of population segments that are not comfortable with the technologies or cannot afford them. For these groups, other, more traditional methods will be required. Even fairly recently, the most effective method of communication and encouraging citizens to evacuate was uniformed officers going door to door (Arlikatti et al., 2006).

3. Zoning

When communicating with the public about who needs to evacuate, it is often useful to divide the study area into geographic zones. Using postal codes or easily identifiable boundaries or landmarks facilitates the identification of threatened homes, compared to risk maps which typically feature topographic and elevation contour information, which has almost no meaning to an average citizen (Arlikatti et al., 2006; Zhang et al., 2004). The number of and size of zones may vary by the type of disaster event. For example, the Houston–Galveston area (Houston–Galveston Area Council, c. 2012) and New York City (ASPCA, c. 2011) have three zones (A–C) designated for hurricane evacuations. Houston–Galveston residents and visitors can find their zip codes on the map and determine their evacuation zones (Houston–Galveston Area Council, c. 2012). Nuclear power plants may have only two zones around them (Physicians for Social Responsibility, 2009), denoted more by circles around the plant rather than conforming to zip codes, or may be further divided in to more easily communicated zones (Rockland County et al., 2002).

Zones are also critical when modeling demand at the aggregate, rather than individual, level. As with urban transportation planning, hurricane evacuation zones are somewhat based on personal judgment (Wilmot and Meduri, 2005). Wilmot and Meduri (2005) reported that technical assistance documents from the 1990s recommended that evacuation zones relate to expected surge flooding, census/traffic analysis zones, and easily identifiable features (e.g. natural features, major roadways) or the zones be otherwise easily communicated, and the number of zones be relatively small per county. They then suggested the addition of uniform elevation and homogeneous land use and the prohibitions of one zone containing another and crossing political boundaries to the zoning considerations.

For no-notice events, zones may be established in the context of phasing, using optimization based techniques (e.g. Hsu and Peeta, 2012) or based on chemical/radiologic threat radii. Wei et al. (2008) suggested defining a harm zone, where citizens are in danger, and a buffer zone where people are not in imminent danger but travel is affected by evacuation activities from the harm zone. Heuristics or assumption based approaches may be used in other contexts, such as flooding (e.g. Wang et al., 2010). Due to the nature of terrorist attacks or transport hazmat spills, it is difficult to establish maps for communicating with the public since the threats can come from any direction and occur at any location. However, such maps may be developed for obvious targets or locations experiencing frequent incidents.

4. Demand modeling

Evacuation demand modeling is complex and depends on a wide variety of factors, observations, and assumptions, as illustrated in this section. Large-scale observations could further enhance understanding of the complexities and help validate the entire demand modeling process.

4.1. Determining who evacuates

The questions of who evacuates and what factors influence the evacuate/stay decision are frequently investigated, mostly in the social sciences, but more recently by engineers as well. Table 1 presents a summary of selected factors influencing this

Table 1

Factors associated with the evacuate/stay decision.

Factor	Likelihood effect	Context								
		General	Hurricane/typhoon	Flooding	Earthquake	Volcano	Fire	Hazmat	Nuclear power plant	Terrorist attack
Previous or recent hazard experience	Mixed (increases and insignificant)		Arlikatti et al. (2006) and Dow and Cutter (1998)	Gruntfest (1977)						
Frequent hazard experience	Decreases		Anderson (1968)	Anderson (1968)		Anderson (1968)				
Previously Evacuated	Increases		Murray-Tuite et al. (2012) and Riad et al. (1999)							
Environmental cues	Increases			Perry (1983)		Perry (1983)		Baron et al. (1988)		
Receiving a warning	Increases		Baker (1979, 1991) and Gladwin and Peacock (1997)	Perry (1981)	Bourque et al. (1971)	Perry (1981)	Fischer III et al. (1995)		Perry (1981)	
Perceived high risk/impact	Increases		Arlikatti et al. (2006), Baker (1979, 1991), and Bateman and Edwards (2002)	Perry (1981)		Perry (1981)			Houts et al. (1984) and Perry (1981)	
Injuries present	Increases									Aguirre et al. (1998)
Distance to threat (closer)	Increases		Carnegie and Deka (2010)		Bourque et al. (1971) and Bourque and Russell (1994)				Cutter and Barnes (1982) and Houts et al. (1984)	
Structural damage	Increases				Bourque et al. (1971) and Bourque and Russell (1994)					
Access to resources	Increases	Vaughan (1995)	Baker, (1979)							
Strong social network	Increases		Baker (1979)	Anderson et al. (1984)						Aguirre et al. (1998)
Older age	Mixed		Gladwin and Peacock (1997) and Riad et al. (1999)						Cutter and Barnes (1982)	Aguirre et al. (1998)
Gender (female)	Increases	Fothergill (1996)	Bateman and Edwards (2002) and Riad et al. (1999)							Aguirre et al. (1998)
Children in the household	Mixed (increases and insignificant)		Baker (1991), Gladwin and Peacock (1997), and Riad et al. (1999)				Fischer III et al. (1995)		Houts et al. (1984) and Perry (1985)	
Race (Caucasian)	Mixed (Increases & insignificant)		Bateman and Edwards (2002), Gladwin and Peacock (1997), and Riad et al. (1999)	Perry and Lindell (1991) and Perry et al. (1982)				Perry and Lindell (1991)		
Income (Higher)	Mixed (Increases & insignificant)		Elliott and Pais (2006) and Gladwin and Peacock (1997)	Perry and Lindell (1991)				Perry and Lindell (1991)		

Table 1 (continued)

Factor	Likelihood effect	Context								
		General	Hurricane/ typhoon	Flooding	Earthquake	Volcano	Fire	Hazmat	Nuclear power plant	Terrorist attack
Residence duration (longer)	Decreases		Baker (1979) and Gladwin and Peacock (1997)							
Pets	Mixed (insignificant and decreases)		Baker (1991) and Solís et al. (2010)							
Work requirements	Decreases		Baker (1979)						Perry (1983)	
Family is together	Increases	Perry (1985) and Quarantelli (1990)								
Fear of looting	Decreases	Quarantelli (1990)							Perry (1983)	

decision. (It is impossible to capture all factors that have been investigated in one table. Readers interested in more details can refer to the individual papers as well as reviews, such as [Sorensen \(2000\)](#).)

Table 1 shows consistency in several factors across disaster types. For example, the risk related factors of environmental cues, receiving a warning/evacuation notice, perceiving high risk or impact (including housing type, living in a surge zone, and strong storms), and proximity to the threat all increase the likelihood of taking protective action regardless of the disaster type. Access to resources to take the protective actions naturally increases the likelihood.

Variables related to experience or exposure to a hazard show inconsistent effects. In the context of hurricanes, [Baker \(1991\)](#) noted inconsistency in the findings for previous experience and length of residence. [Arlkatti et al. \(2006\)](#) supported Baker's conclusion that how experience is measured and how residents interpreted the experience determine the effect of these variables.

Many of the socio-demographic and economic characteristics show varied effects between insignificance and significance. Except for the older age or presence of elderly, when significant, the direction of significance is consistent. One characteristic that is consistently significant is gender (female), which is associated with higher risk perception.

Most of the studies identifying the influence of variables on the decision to evacuate or take other protective action used frequency of response and logistic regression techniques. A more recent technique, illustrated in [Hasan et al. \(2011\)](#), is the mixed logit, or random parameters logit model. This type of model allows for heterogeneous effects for selected variables. [Hasan et al. \(2011\)](#) found heterogeneities in the effects of voluntary evacuation notice, work requirements, and the number of children. The heterogeneities with respect to children further support the mixed effects shown in Table 1.

4.1.1. Influence of neighbors and social networks

As shown in Table 1, one of the factors influencing the decision to evacuate or not is the strength of the social network. Friends and relatives may be the first sources of information about a threat and/or advisories to evacuate ([Lindell and Perry, 1987](#)). Contact with friends and relatives can convince people that a situation is dangerous or their appeals to evacuate may lead to the desire to appease them even if the evacuee's initial response was to remain ([Baker, 1991](#)). The response of family and neighbors in the social network can encourage a given individual/household to evacuate; however there is a high correlation of authorities' efforts in certain areas and subsequent observations of business closures and evacuations. This correlation leads to difficulty in determining whether the official warnings (and residents are leaving for the same reasons) or the social cues (conformity effect) are the causal factor ([Baker, 1991](#)). Presuming the social cues as the causal factor, [Hasan and Ukkusuri \(2011\)](#) created a social contagion model to investigate the conditions for a cascade through the network of the decision to evacuate. They also considered the effects of social community mixing patterns, first person to decide to evacuate, and decisions neighbors made in previous time steps. Their model provides a mathematical and simulation approach to investigate these factors in bringing about the desired behavior of evacuating an area based on social ties. An earlier combined diffusion and contagion model was developed by [Rogers and Sorensen \(1991\)](#) for the spread of emergency warnings. Their model uses a logistic function to represent the diffusion of emergency warnings. The parameters on this model can be adjusted to represent different warning systems based on the maximum penetration of each system, which is related to the characteristics of the system and the likelihood of the public being in different locations at a particular time. The contagion aspect of the model represents people first hearing of the event and spreading the information to others ([Rogers and Sorensen, 1991](#)).

4.2. Trip generation

Trip generation approaches generally attempt to capture some of the social and threat factors influencing the decision to evacuate with the goal of producing time dependent demand for traffic simulation. Evacuation trip generation has been per-

formed with a variety of techniques, several of which are discussed below. As noted in Wilmot and Mei (2004) and Pel et al. (2012), time dependent demand can be determined in two steps or one.

4.2.1. The two step approach

If the two step approach is adopted, the first step involves estimating the number of evacuating households. The second step of the two step approach determines the departure time often with a response or mobilization curve (Pel et al., 2012; Wilmot and Mei, 2004).

4.2.1.1. Number of evacuating households. This first step may involve participation rates, which are similar to cross-classification techniques (Wilmot and Mei, 2004) and the most frequently used approach (Fu and Wilmot, 2007). Rates are specified for each zone and multiplied by the zone's population to estimate the total evacuees (Wilmot and Mei, 2004). Actual evacuation rates may vary across disaster types and due to unique event characteristics (Drabek, 1986).

Another approach is logistic regression, as in the models determining the factors influencing the decision to evacuate. The logistic regression models predict a binary outcome – to evacuate or stay. Logistic regression and participation rate techniques were compared with neural network approaches in the hurricane context by Wilmot and Mei (2004). They concluded that the same variables used in logistic regression could be used in the neural networks for comparison purposes. Results of the comparison indicated that logistic regression and neural network models were more accurate than the participation rate model at both the aggregate and disaggregate levels and provided a lower overestimation rate than the participation model. Wilmot and Mei (2004) did not find a clear preference for either the logistic regression or neural network models in terms of demand accuracy, although they noted marginally better performance of the neural network models. Pel et al. (2012) indicated that neural network approaches, as well as other data driven models, are calibrated with specific data and the resulting models may not be transferable.

Hasan et al. (2012) investigated the issue of transferability of binary logit models among three hurricanes (Andrew, Ivan, and Katrina) for the decision to evacuate or stay. They estimated separate as well as joint models and used the likelihood ratio test specified in Washington et al. (2011) to test the transferability. They found transferability to exist but results have to be interpreted carefully.

4.2.1.2. Departure time. With total demand estimated, evacuees must then be distributed over time. The evacuation departure process is particularly important in the modeling of network traffic processes. Early approximations of the traffic loading curves were based on empirical evidence, stated intention surveys, planner judgment, and simulation of the warning message diffusion (Southworth, 1991). The time spent responding to a warning often follows an S-curve (Sorensen, 2000). Similarly, many evacuation models have been based on cumulative departure S-curves, most notably the long established US Army Corps of Engineers response rate curves such as the example shown in Fig. 1 (US Army Corps of Engineers, 1999). These curves, which also include the actual response observed during Hurricane Opal, follow a pattern of response that has been repeatedly observed over numerous hurricane evacuation events. In traffic models, they have been translated into mathematical formulations using logistic functions to form sigmoid curves from which evacuation traffic demand can be computed for individual time periods.

Under the S-curve assumptions, about 10% of the total evacuating population departs prior to any official evacuation order (at time = 0). Then, based on the urgency of the order and the response it generates, the curve slope increases as more departures occur. Fig. 1 includes three different curves that represent movements under more pressing levels of urgency such as may occur under hazards of more immediate threat or for orders issued at a different time of day.

Continued study and observation have shown these curve shapes to be generally applicable for other types of hazards that may give varying amounts of advanced warning time or may incorporate faster moving threat conditions such as recent studies by Cova et al. (2011), Dennison et al. (2007), and Wolshon and Marchive (2007) for wildfire evacuations. The main difference, however, would be the temporal scale of the x-axis as well as the slope of the curve. Interestingly, the analysis of traffic count data from Southeast Louisiana observed during the evacuation for Hurricane Katrina in 2005 showed a similar departure curve pattern, although, in that event, there were back-to-back S-curves representing the cumulative traffic over the 2-day period of that evacuation (Dixit et al., 2011a) as shown in Fig. 2. Such back-to-back S-curves occur due to high demand in the morning that tapers off throughout the day. In hurricane evacuations, there are usually several days of forewarning and people prefer to leave earlier in the day so they can reach their destinations before nightfall (Lindell et al., 2005).

While early surveys involving departure times were not designed for detailed traffic modeling and lacked spatial resolution (Southworth, 1991), advances have been made over the last decade. Fu et al. (2007) sought to develop a hurricane evacuation response curve that was responsive to certain influential characteristics of the hurricane, time of day, and the type and timing of evacuation notice issued. They developed a model based on datasets from Hurricanes Floyd and Andrew to test a series of storm scenarios and different evacuation notice policies. The model predicted responses that were similar to observed behavior for similar storm conditions. It was thought that because the model included evacuation notice as an independent variable, it would also serve as a useful tool to analyze evacuees' responses to different evacuation policy decisions, such as the type and timing of an evacuation notice under different storm conditions.

As an example of the practical impact of varying the response model on an evacuation, Tamminga et al. (2011) examined the impacts of departure duration on total evacuation time and roadway network performance in The Netherlands. Among the findings of the research was that the outflow from an evacuation network remained constant because the number of

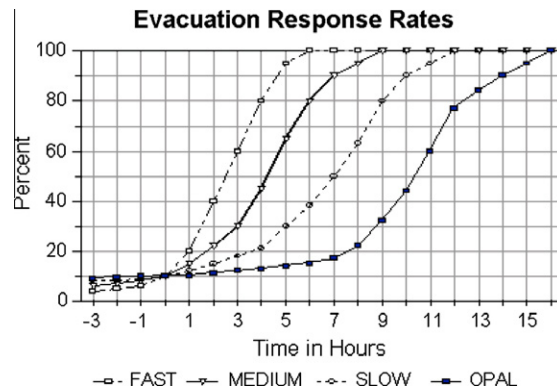


Fig. 1. Evacuation behavioral response curves. Source: US Army Corps of Engineers.

departures from a protective action zone is a function of the capacity of local bottlenecks upstream of the destinations. As the number of vehicles increases, there are corresponding increases in congestion. Thus, for increasingly shorter evacuation durations that tend to result in more abrupt surges in demand, such as those that might occur during the most urgent conditions, internal network gridlock results and lengthy delays are incurred by evacuees who are unable to leave the network. Conversely, increases in demand that permit high but orderly flow can result in greater numbers of vehicles clearing the network in shorter time periods.

While the above scenarios describe evacuation processes that are limited by capacity of the network, portions of the Fig. 1 curve occur due to other conditions. In particular, the upper-right segments of the curves where the response data points become asymptotical to the 100% participation line can be described as the “demand-limited” portion of the response process. Reasons for this condition vary, but they can occur in more sparsely populated areas where there is not overwhelming evacuation demand and evacuees feel confident in leaving late and for evacuees that need additional time to make last minute departure preparations, e.g., tend to friends, relatives, pets, etc. or otherwise shut down agricultural and industrial operations. These phenomena have been described in several past works including Lindell et al. (2002) and then later by Wolshon et al. (2010) where it was discussed as the “evacuation tail” during evacuations for nuclear power plant emergencies.

4.2.2. The one step approach

Used less frequently than the two step approach, the one step approach simultaneously determines the number of evacuating households and departure times. The technique used is the repeated application of a binary logit model. At each iteration of the model, the outcome is to evacuate now or stay/postpone the decision. This technique appears to have been first utilized for hurricanes by Fu and Wilmot (2004) who were able to reasonably replicate the departures by time of day across the 3 days before Hurricane Andrew’s landfall. They later tested the transferability of the model developed for Hurricane Floyd with the Hurricane Andrew data and found that the two data sets were not completely compatible and some error was introduced with the way distance was transformed for the model (Fu et al., 2006). In a comparison of the sequential

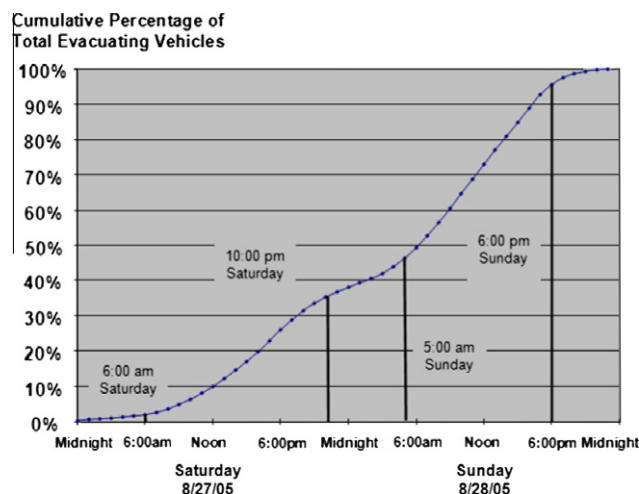


Fig. 2. Two-day cumulative evacuation double-S response curve, southeast Louisiana Hurricane Katrina evacuation. Data source: Louisiana Department of Transportation and Development.

logit model with the participation rate with response curve approach, [Fu and Wilmot \(2007\)](#) concluded that the sequential logit model was preferable for hurricane evacuations.

Also in the context of hurricanes, [Yazici and Ozbay \(2008\)](#) compared the sequential logit model with S-curves and the Rayleigh distribution for traffic loading. They found that these models produced different demand generation and loading patterns. Based on a small, simplified network, they examined the network performance under the different patterns and found the performance varied based on the loading curve. They concluded that the models must be selected and calibrated carefully.

4.2.3. Additional trip considerations

While the bulk of evacuation trip modeling has focused on the evacuation trip itself for those at risk, other trip types, such as background traffic, intermediate trips, and shadow evacuation ([Zeigler et al., 1981](#)), play important roles in measuring the network performance and estimating evacuation time.

4.2.3.1. Background traffic. Background traffic refers to vehicles in the network that are not part of the active evacuation. Such traffic is often estimated as a percentage of normal (i.e., non-evacuation) trip tables or a modification thereof (e.g. [Brown et al., 2009](#); [Paz de Araujo et al., 2011](#)). [Zheng et al. \(2010\)](#) separated background traffic into two categories: (1) those that are not affected by the disaster but may be affected by traffic control measures and broadcast information and (2) those that have an origin and/or destination affected by the threat. In the latter category, trips with threatened origins become evacuees while trips with threatened destinations require new destinations. In general, little information is available about the amount of background traffic that would be present during an evacuation (e.g. [McGhee and Grimes, 2006](#)).

4.2.3.2. Intermediate trips. Some of what might be otherwise classified as background traffic is actually an *intermediate trip* – a trip undertaken by evacuees but not the final departure to the ultimate destination. Based on social scientists' observations that families gather and evacuate as a unit (e.g., [Stern, 1989](#)), [Murray-Tuite and Mahmassani \(2003, 2004\)](#) reasoned that this behavior should be incorporated into evacuation traffic simulations, at least for no-notice events. They used optimization techniques to sequentially identify optimal meeting locations and pick up assignments for drivers to gather dependent family members. The resulting trip chains served as input to traffic simulation. This activity based concept was also investigated by [Lin et al. \(2009\)](#) and [Ma et al. \(2009\)](#) allowed meeting at home.

In a later study involving modeling behavior based on information collected from interviews, [Liu et al. \(2012\)](#) developed binary logit models to determine the factors influencing child pick up expectations during emergencies as well as daily travel. They found both mothers and fathers more likely to gather children in an emergency than on a normal day. In emergencies, the distance between a father and a child had no effect on the father's probability of picking up the child, but for mothers, the probability decreased ([Liu et al., 2012](#)). Liu further developed statistical models for spouse pickups and meeting at home. She then used the models with mode selection and traffic simulation tools to find the impact of considering the gathering behavior, discovering significantly longer network clearance times ([Liu, 2011](#)), supporting [Murray-Tuite and Mahmassani's \(2004\)](#) previous work. [Murray-Tuite et al. \(in press\)](#) compared the activity chains reported in the interviews to the results of an optimization model that simultaneously determined optimal meeting locations, destinations, modes, and pick-up assignments and sequences. They discovered that many households expected to conduct pickups according to normal responsibilities and preferred to meet at home; these choices were only optimal in some cases. Greater deviations from optimality arose when the household member dispersion over the network, and hence, the complexity of evacuation logistics, increased ([Murray-Tuite et al., in press](#)).

4.2.3.3. Shadow evacuation. Shadow ([Zeigler et al., 1981](#)) evacuation refers to the movement of evacuees who are not part of the required evacuation. Numerous issues associated with shadow evacuation can affect the evacuation process. In some areas, an over-evacuation of people not threatened by the direct effects of a hazard can create additional traffic and congestion within a network and prohibit the movement of evacuees who are directly threatened. This has been a planning concern in the Houston Metropolitan Region where shadow evacuation can greatly restrict the movement of evacuees from high risk areas, such as Galveston Island. In fact, this particular problem was highlighted by conditions in Hurricane Rita. Looking back at [Table 1](#), one can infer that many shadow evacuees perceived high risk to themselves if they remained where they were. Although the concept of shadow evacuation has been researched to some degree as discussed by [Sorensen and Vogt \(2006\)](#), few studies have investigated the issue of shadow evacuation separately from the necessary evacuation. Modeling techniques would be similar to those already outlined above.

4.3. Trip distribution and destination choice

In the context of evacuations, destination choice can pertain to accommodation type as well as the location of the accommodation. Types have generally been classified as friends'/relatives' homes, hotels/motels, and shelters, ordered from most to least commonly preferred ([Lindell et al., 2011](#); [Wu et al., 2012](#)). [Sorensen \(2000\)](#) indicated that shelter use averages about 15% among the evacuees. Accommodation type is often modeled through discrete choice approaches, such as multinomial logit models. As noted in [Murray-Tuite et al. \(2012\)](#), variables influencing the choice include hazard severity, income ([U.S. Army Corps of Engineers, 2001](#)), size of the evacuation, type of emergency, time of the evacuation, age, and ethnicity ([Mileti et al., 1992](#)). To this list, [Whitehead et al. \(2000\)](#) add education and pet ownership.

The geographical locations of the accommodations selected are often obtained from surveys of evacuees. Discrete choice models are infrequently developed due to the size of the choice set. Cheng et al. (2008) represents one exception for a hurricane in South Carolina. They reduced the number of destination choices to 28 and used a multinomial logit model to predict the zone selected as a destination. They developed separate models for the different accommodation types friends'/relatives' homes and hotels/motels. Significant variables included in the modeling were distance to the destination, population at the destination (for the friends'/relatives' homes model), number of hotels/motels (for that model), an indicator of whether the destination zone experiences high wind, percentage of the destination population that was White, and an indicator of whether the destination was in a metropolitan statistical area, and an indicator of whether the zone contained an interstate highway. Their models fit the observed data reasonably well (Cheng et al., 2008). Using the same survey data, they later extended this work to include time-dependent travel cost rather than the previous distance consideration. They found the previous static model to slightly outperform the model with dynamic cost but suggested further investigation into other modeling approaches, such as nested logit models of time period and destination (Cheng and Wilmot, 2009).

More often, subjective judgment (Cheng and Wilmot, 2009) or aggregate approaches similar to those used in normal transportation planning are used to extend the destination choice to the larger evacuating population. Work in this aggregate area has been relatively limited to date, with most of the existing work performed by Wilmot and colleagues in the context of hurricanes (e.g., Chen, 2005; Cheng et al., 2011; Wilmot et al., 2006). Using the same Hurricane Floyd data as in the discrete choice work, Wilmot et al. (2006) compared the static intervening opportunity and gravity models with separate models for the friends'/relatives' homes and hotels/motels accommodation types. They found only small differences between the gravity models, intervening opportunity models, and an extended intervening opportunity model that accounted for the evacuation direction relative to the storm path. Within the small performance differences, the extended intervening opportunity model was the best, followed by the gravity models. They then tested the transferability of the gravity model developed for Hurricane Floyd in South Carolina to Hurricane Andrew in Louisiana and found no significant difference in trip length frequency diagrams or sets of friction factors (Wilmot et al., 2006).

Cheng et al. (2011) then investigated a dynamic gravity model with the Hurricane Floyd data. The dynamic impedance function considered the projected path of the storm as well as network conditions and accommodation availability. They concluded the model results were satisfactory but suggested that further tests of the transferability of the combined dynamic demand generation and dynamic gravity models remains to be tested (Cheng et al., 2011). Pel et al. (2012) further suggested using other traffic assignment tools to compute the network conditions and travel times.

For evacuations involving less notice and planning, one may consider a separation of *ultimate* from *proximate* destinations. As discussed in Lindell and Prater (2007), *ultimate* destinations are where the evacuees will stay until permitted to return to their homes while *proximate* destinations are the points at which the vehicles leave the risk area. The previously discussed destination choices are considered ultimate destinations. Lindell and Prater (2007) indicated that *proximate* destinations are related to route choice and cited Barrett et al. (2000) as identifying the nearest safe destination, temporally closest safe destination, and minimum perceived cost safe destination as possible proximate destination choices. However, these considerations were not data driven (Lindell and Prater, 2007).

4.4. Mode split/choice

The mode of transportation taken to evacuate is likely to depend on a number of factors, including, but not limited to, (1) characteristics of the disaster, (2) required travel distance to reach safety, (3) location of the evacuees at the time an order is given, and (4) available options. These considerations may be interdependent. For example, 9/11 occurred with no-notice, people were uncertain about the distance required to reach safety, many people were at work, and available travel options were dynamic and differed from a typical day (e.g., airports, bridges, and transit were shut down while boats came to rescue people from Manhattan). Compare 9/11 to a hurricane evacuation, which often comes with at least a full day of notice, may require a greater travel distance, people may choose to depart from work or home, and to the degree available, may choose air, transit, or the more likely car or carpool.

Identifying which modes would be available for a particular disaster from a physical perspective as well as a service perspective is complex, suggesting that a full understanding of the effects of the disaster on physical infrastructure, security protocols, and service provider capabilities is needed. The social aspects of the willingness of drivers/pilots to operate under the particular disaster conditions should also be accounted for. Then, the more traditional aspects of mode choice/split can be considered.

Mode split had been relatively understudied until Hurricane Katrina, although mass transit has been an element of many previous evacuations (e.g., the Three Mile Island evacuation (Chenault et al., 2002)). Since then, several researchers (e.g., Renee et al., 2009, 2008) have investigated the needs for the carless and special needs populations. These segments of the population are dependent on others for transportation (e.g., a transit/emergency management agency, family, friends, or neighbors). Wu et al. (2012) found only 11% of evacuees not taking their own cars, of which "71% rode with someone else and 28% used another form of transportation" (pp. 450–451). Wu et al.'s summary of findings on previous transit use rates implied a top value of 13%. As Lindell and Prater (2007) pointed out, transit operations may require turns, affecting contra-flow strategies, reduce outbound capacity, and may cross evacuation routes.

Those who are able to take personal vehicles are still largely assumed to take them, both from the citizen and management agency perspectives. In one of the few discrete choice studies on emergency mode choice, Deka and Carnegie (2010)

developed models based on stated preference household survey data that revealed a strong preference for private vehicle (84%) and that the choice of other modes was related to familiarity with a particular transit option and the unavailability of a personal vehicle.

For households with access to at least one personal vehicle, relevant questions are how many the household will take and what types of vehicles they are. Lindell and Prater (2007) summarized earlier works' assumptions on the number of vehicles as all registered vehicles and Poisson distribution estimates (with mean of 1). They further indicated that behavioral expectations reported a range of household averages between 1.10 and 2.15 (Lindell and Prater, 2007). The size and characteristics of the personal vehicles can have capacity effects for the network. For instance, Lindell and Prater (2007) recommend accounting for cars pulling boat trailers in hurricane evacuation modeling. Similarly, trailers containing livestock, recreational vehicles, and trucks can lower outbound capacity. The vehicle type capacity reductions have rarely been accounted for in evacuation simulations.

Other items to consider when predicting modes of transportation are different population groups. Tourists, for example, may be more likely to fly back home than evacuees are to fly away from home. People conducting intermediate trips (e.g. picking up children (Lin et al., 2009; Liu et al., 2012; Murray-Tuite et al., in press; Murray-Tuite and Mahmassani, 2003, 2004); returning home (Liu, 2011; Ma et al., 2009)) may use different modes for different legs of their trips, although one must be careful not to assign a personal vehicle to a leg unless a personal vehicle is taken on the previous leg or available at the intermediate destination (Murray-Tuite et al., in press). Finally, other background traffic is usually assumed to behave according to a normal day.

5. Traffic assignment and route selection

Since transit services are anticipated to follow pre-defined routes, we here focus on associating drivers with routes. This association can be done at the individual driver or aggregate level.

5.1. Individual perspective

Prior to the recent advances in mobile data collection technology (e.g. Blue Tooth, smart phones), determining which routes drivers selected and/or why involved surveys or interviews, and to capture population segments without advanced technologies, this technique will still be required in the future. Most of the previous surveys on route choice and reasons for selection pertained to hurricanes (e.g. Dow and Cutter, 2002; Murray-Tuite et al., 2012). Murray-Tuite et al. (2012) found that evacuees most often seek familiar routes, which echoes previous findings (e.g. Lindell and Prater, 2007; Vogt and Sorensen, 1992). Dow and Cutter (2002) found strong preferences for interstate routes that was at least partially attributed to concerns about services, isolation on rural roads in cases of emergencies, and cell phone coverage. The implications of their findings included heavy congestion on interstate routes and underutilized alternative routes.

5.2. Traffic assignment tools

At the aggregate level and to answer questions about evacuation times, one can use a variety of traffic assignment-simulation tools to move given demand over space, including, but not limited to, Paramics (Chen and Zhan, 2004; Cova and Johnson, 2002), Vissim/Visum (Edara et al., 2009; Elmitiny et al., 2007), DynusT and DYNASMART (Chiu et al., 2005; Murray-Tuite and Mahmassani, 2004; Sbayed and Mahmassani, 2006), DynaMIT (Balakrishna et al., 2008), NETVAC (Sheffi et al., 1981), DYNEV (KLD Associates, 1984), MASSVAC (Hobeika and Jamei, 1985), CORSIM (Gu, 2004; Lim and Wolshon, 2005; Sisiopiku et al., 2004; Theodoulou and Wolshon, 2004b), MATSim (Lämmel et al., 2010), TRANSIMS (Naghawi and Wolshon, 2011a; Wolshon et al., 2009), and CEMPS (Pidd et al., 1996). With the exceptions of CEMPS, NETVAC, and MASSVAC, these tools are also used in non-evacuation planning. Pel et al. (2012) and Alsnih and Stopher (2004) provide historical perspectives and summaries of these and other traffic simulation tools for evacuations.

A limitation with simulation tools is that they rely on pre-specified rules of behavior that are difficult to validate for evacuations. Lindell and Prater (2007) and Pel et al. (2012) criticize the use of user equilibrium for evacuations due to the inability of evacuees to learn traffic conditions by experience since the evacuations are rare events. The routing behavior assumptions should be further explored through interactions of social science and engineering based studies that focus on the individual with rule based simulation model studies.

Pel et al. (2012) also include a discussion of en-route route choice models, where drivers make decisions at each intersection and hybrid route choice models, where some drivers follow one set of rules (e.g., historical paths) and others follow another set (e.g., en-route switching). They found few examples of evacuation studies with en-route choice models, although NETVAC allows myopic route improvisation (Lindell and Prater, 2007). The hybrid route choice models include, but are not limited to DYNASMART, DynusT, and EVAQ (used in (Pel et al., 2008, 2010)). The first two models are mesoscopic while the last is a macroscopic model. See Pel et al. (2012) for additional discussion of traffic assignment modeling.

5.2.1. Evacuation flow rates and models

As more traffic data during evacuations were collected and analyzed, persistent trends have begun to emerge to suggest that traffic flow, particularly maximum flows, under evacuation conditions differ from those of non-emergency periods

(Wolshon and McArdle, 2009). The importance of this finding may be significant in the modeling of evacuations since most of the performance of macro-level simulations is based on the relationship of demand to capacity. Because of the consistent and compelling evidence of the flow rate variation, the Florida Department of Transportation recently recommended the use of Maximum Sustainable Evacuation Traffic Flow Rates (MSETFR) for evacuation traffic simulation modeling in the Florida Keys. Evacuations are particularly critical in this chain of islands because there is only a single route of egress for over 80,000 residents and visitors. Additional research to provide a quantitative basis and explanation of evacuation flow phenomena was conducted by Dixit et al. (2011b).

5.3. Multi-mode simulation

With the increased interest in addressing mode choice/split, the traffic simulation models need to account for transit modes as well as cars. Transit modeling capabilities are present in most of the tools listed above, including Paramics, Visum/Visum, DynusT, DYNASMART, TRANSIMS, VISTA, and MATSim. The evacuation studies that explicitly deal with multiple modes of transportation are relatively limited before Hurricane Katrina.

When dealing with optimization for transit routes and stops, the transit features of these tools are used. For example, Abdelgawad and Abdulhai (2010) use DynusT in their optimization of transit routes and schedules and optimization of personal vehicle schedules, destinations, and routes. Although not clear about which traffic simulation model was used, Chen and Chou (2009) used simulation of traffic and traffic control when optimizing transit operations for evacuation. Mastrog-iannidou et al. (2009) used VISTA in their emergency transit vehicle assignment. Naghawi and Wolshon (2010) used TRANSIMS to evaluate different transit based strategies, finding that the most effective scenarios involved use of transit during auto-based evacuation lulls. In subsequent work, they found that buses minimally impacted arterial evacuation routes but caused additional delay on freeways (Naghawi and Wolshon, 2011b).

Elmitiny et al. (2007) used VISSIM to evaluate the evacuation of transit facilities. They also captured the effect of the evacuation on the traffic network. Also using VISSIM's transit capability, Liu et al. (2011) optimized the relocation of children for pickups, allowing different modes of transportation.

Optimization based planning seems to form the bulk of the literature pertaining to multi-mode evacuation simulation although a few heuristic or judgment based plans have also been tested. Testing the plans for a real event remains a future possibility.

5.4. Dynamic decisions and information

With the capability of many tools to simulate information provision through variable (dynamic) message signs and/or enroute information through other devices, some studies include these strategies as part of the larger study, while others focused on these issues in isolation. For example, Robinson and Khattak (2010) conducted a behavioral intention survey of Hampton Roads area residents to identify frequently used information sources and variables associated with route changing. In terms of information sources while driving, they found (in order of highest to lowest use) radio, advanced traveler information systems (ATIS), phone, internet, and GPS devices were substantially used. With information provided by ATIS or radio, depending on the congestion period, the percent of drivers expecting to divert ranged from 32.5 (0 min of congestion) to 93.8 (120 min of congestion). They later combined the results of this survey with traffic simulation to examine the effectiveness of ATIS under incident conditions during an evacuation (Robinson and Khattak, 2012).

Through simulation with EVAQ, Pel (2011) investigated the impacts of different sensitivities to route updating thresholds and how well informed travelers are on evacuation times and network exit point usage. As expected, when travelers are not well informed and are unfamiliar with the network, exit point use is less efficient and travel times are higher. However, he notes that raising the information level past a certain point causes deterioration in network performance (Pel, 2011), which is consistent with previous studies of information provision.

6. Evacuation strategies

Recognizing that evacuations will generally generate severe congestion at the time when lives are threatened, many researchers and agencies have investigated demand and/or supply strategies to reduce evacuation time. Abdelgawad and Abdulhai (2009a) list contraflow operations, modified traffic control, route guidance, and staged evacuation as the popular strategies. These strategies and others are discussed below.

6.1. Supply side

Supply side strategies include contraflow operations, crossing elimination, special signal timings, transit operations, and shoulder lane use.

6.1.1. Contraflow

The use of contraflow traffic operations during evacuations increased significantly after Hurricane Floyd in 1999 when it was implemented for the first time on a significant scale. Contraflow is a form of reversible traffic operation in which one or

more travel lanes of a divided highway are used for the movement of traffic in the opposing direction ([American Association of State Highway and Transportation Officials, 2004](#)). It is a highly effective strategy because it can both immediately and significantly increase the directional capacity of a roadway without the time or cost required to plan, design, and construct additional lanes. It is also popular with the public because it is viewed as a logical utilization of the unused lane capacity of adjacent inbound lanes.

Today, contraflow is one of the most widely used traffic management techniques in states threatened by hurricanes and has been developed for use in coastal states from New Jersey to Florida on the Atlantic seaboard and from Florida west through Texas along the Gulf of Mexico ([Wolshon, 2001](#)). The effectiveness of contraflow during a live operation was quantified for the first time by [Wolshon \(2008\)](#) based on traffic counts recorded during the Hurricane Katrina evacuation of south Louisiana in 2005. The flow rates measured during this event were about 75% of the adjacent normally flowing lanes. Although no firm explanation for these lower rates has been determined, this reduced flow is consistent with modeling predictions ([Post Buckley Schuh and Jernigan Inc., 2000](#)) and simulation studies ([Lim and Wolshon, 2005](#); [Theodoulou and Wolshon, 2004a](#); [Williams et al., 2007](#)).

6.1.2. Crossing elimination

Another traffic management strategy that has been suggested to facilitate the movement of traffic during urban evacuations is the elimination of certain turning and crossing maneuvers at intersections. Although somewhat similar, this area of work is not nearly as well developed as modeling for emergency network routing. The concept of movement prohibition is to remove conflicting traffic stream interruptions in favor of continuous flow in the primary outbound direction away from a hazard. This would be accomplished using police control or by placing barricades in intersections. [Cova and Johnson \(2003\)](#) presented a network flow model for identifying optimal lane-based evacuation routing plans within a complex urban road network. Based on an integer extension of the minimum cost flow problem, it was used to generate routing plans that trade total vehicle travel-distance against merging, while preventing traffic crossing-conflicts at intersections. These plans were then evaluated using capacity analysis and microscopic traffic simulation techniques. The results of these analyses showed up to a 40% reduction in travel time depending on the network configuration and hazard scenario. Cova and Johnson's general idea has recently been expanded by several other researchers, such as Xie and colleagues ([Xie and Turnquist, 2011](#); [Xie et al., 2011](#)).

6.1.3. Special signal timings

Recent experiences have shown that the progression of evacuation traffic can be significantly impacted by traffic signals. While this is particularly true for evacuations in urbanized areas, traffic signal delay can also affect flow in rural areas as evacuating traffic streams pass through periodic signalized intersections between their origins and destinations. Since the majority of model development efforts have historically been focused on freeways, the analysis of the impact of signal timing on evacuation has been less studied. Recently, however, this has begun to change. Several studies, including studies by [Chen et al. \(2007\)](#) and [Parr and Kaiser \(2011\)](#) have examined strategies for timing and coordinating signals in Washington, DC for no-advance-notice disasters. The former study used a CORSIM simulation model to assess the trade-offs between evacuation clearance time and average delay under different timing plans on a primary route within the city.

In addition to modeling work that looked specifically at signal timing and coordination, another branch of recent modeling work has evolved to examine network control strategies. Work by [Liu et al. \(2007\)](#) involved the development of a model reference adaptive control framework for real-time traffic management under emergency evacuation. The idea of the system was to be able to dynamically control traffic flow under evacuation conditions in such a way that the overall performance of the system could be improved during the emergency by using prevailing traffic conditions to produce traffic control schemes in real-time. Simulation studies done in conjunction with the project showed it could significantly improve the performance evacuation traffic networks.

6.1.4. Transit

In the wake of Hurricanes Katrina and Rita, the planning of mass transportation systems to serve mobility limited populations during emergencies became an area of emphasis for municipalities across the US. Similarly, the modeling and analysis of transit-based evacuations became an area of research emphasis. Over the past half-decade, transit operations have been analyzed using modeling to optimize scheduling and operations (([Abdelgawad et al., 2010a,b](#); [Bish, 2011](#); [He et al., 2009](#))). Additional modeling studies of transit bus evacuation operations were conducted by [Naghawi and Wolshon \(2010, 2012, in press\)](#). In these models, the Citizen Assisted Evacuation Plan for metropolitan New Orleans was evaluated to assess bus travel time and operating speeds under alternative evacuation routing plans and under a range of response urgencies.

6.1.5. Use of shoulders

In addition to capacity enhancements like contraflow and improved signal timing, the use of shoulders on freeway evacuation routes has also been planned. The idea of using shoulders for an additional travel lane is not new. The Virginia DOT has used shoulder lanes for many years to gain additional capacity during rush hours in the suburban Washington, DC area ([Wolshon and Lambert, 2004](#)). Shoulder lanes were also suggested, though never implemented for hurricane evacuations in the State of Florida. After the Hurricane Rita evacuation of 2005, the Texas Department of Transportation (TxDOT) conducted

extensive studies of ways in which they could learn from the shortcomings of that event. Among the recommendations of this study was the use of shoulders as “Evaculanes” on the outside shoulder of both the normal and contraflow sides of free-ways. A report by Ballard and Borchardt (2006) included the basis for the development of these recommendations, including the conversion of shoulders into evacuation lanes. A follow up study by Ullman et al. (2007) used focus group input and surveys to develop guidelines for special use hurricane evacuation traffic control devices, including shoulder lane signs and pavement markings.

6.2. Demand side

An alternative or complement to increasing the evacuation route supply is the management of demand. This can be done from a variety of perspectives: minimizing shadow evacuation and background traffic, sheltering in place, and phased (staged, sequenced) evacuation. Reducing shadow evacuation is likely to be best accomplished through clear communication and education. Since the decision to evacuate is largely governed by feeling unsafe, convincing people to remain rather than evacuate involves making them feel safer where they are rather than on the road for evacuation purposes. Similarly, sheltering-in-place or close by is a safer course of action for some disaster types (e.g. tornados) and this needs to be clearly communicated to the public prior to and when such an event occurs. Reducing background traffic also involves communication so that people who do not need to be on the road cancel some of their trips; however, some background traffic is still likely to occur since those not involved in the evacuation will have work and family responsibilities. Since these demand considerations are not directly under the control of transportation agencies, they are not often explored in the transportation literature.

On the other hand, evacuation staging has received a moderate amount of attention. The basic idea of this strategy is to find an optimal sequence of evacuation assignments to minimize evacuation time, risk, or the time for those in the highest risk areas to reach safety. One of the first studies in this area involved comparing staging and simultaneous evacuation of three network structures; the number of zones for sequencing was set to four (Chen and Zhan, 2004). Since then, the number of zones considered has grown and approaches have tended toward complex optimization techniques. For instance, Sbayti and Mahmassani (2006) developed a bi-level formation that determines time-dependent route assignments at the upper level with the method of successive averages and the route travel times at the lower level, solved with traffic assignment-simulation software. The modeling process assigns each vehicle a departure time, route, and destination. They used this output to generate a time-dependent staging plan for each origin. Similar to Sbayti and Mahmassani, Bish et al. (2006) developed an optimization model to determine evacuees' destinations and departure times. Also determining the optimal destinations and scheduling of evacuation demand, Abdelgawad and Abdulhai (2009b) used evolutionary algorithms to interact an optimization model with dynamic traffic assignment (to perform the routing). Chiu et al.'s (2006) model also determined optimal destinations, routes, and stages. Their model was based on the cell transmission model and minimized travel time for all evacuees in the network. Also based on the cell transmission model, Dixit and Radwan (2009) examined the sequencing of evacuation orders for cities in a hurricane evacuation, incorporating a sequential logit demand model. The sources agree that staged evacuation reduces evacuation times; however, citizen (and even local emergency managers and elected officials) compliance with this strategy remains to be gauged. Lower than anticipated compliance will affect the performance of this strategy and field tests are needed to assess the impact of compliance levels.

In addition to the optimization models involving departure times listed above, several researchers developed optimization approaches that combined destinations and routes, which can be considered part of the determination of proximate destinations. For instance, Han et al. (2006) sought the minimum user optimal or system optimal travel time and used the technique of connecting all true network destinations to a single super-destination with dummy links. They determined the real destination by tracing the flow on these dummy links. As with the staged evacuation, compliance with assigned destinations and routes needs to be evaluated.

7. Special considerations and logistical difficulties

Similar to the earlier discussion of the use of mass transit resources for the evacuation of carless populations, the planning of evacuations for special care facilities like hospitals gained significantly increased attention in the wake of Hurricane Katrina. In most locations, responsibility for the movement of individuals who reside in care facilities fell to the manager of that facility. For example, care centers like hospitals and senior residential areas signed cooperative agreement with facilities elsewhere to provide shelter for patients and residents. Transportation between these locations was contracted to various providers like ambulance and bus companies. After closer inspection after Katrina, it was found that, in some cases, multiple facilities were contracted with one or two providers who could deliver services to any one or several of the facilities, but who could not provide service to all of them simultaneously such as might be required during a major emergency like a hurricane or earthquake.

The Katrina disaster and the evacuation behaviors and tendencies that it revealed also prompted action by the United States Nuclear Regulatory Commission (NRC). As the licensing body for nuclear power plants in the US, the NRC has keen interest to maintain a high level of awareness in emergency preparedness and behavioral response, particularly as they relate to evacuations. While many of the results of NRC studies do not appear in the scholarly literature, they are among the

leaders in supporting such studies; several of which are referenced here. The following sections present findings of recent reviews and research on the movement of evacuees from schools and special care facilities as well such as those published by the NRC as plans and research conducted elsewhere on the provision of services to evacuees while en route to their destinations.

7.1. Hospitals and special facilities

McGlown (2001) compared two models from studies related to the evacuation process to examine variables perceived by hospital administrators as critical in their decision to evacuate their facility compared to variables identified in the literature. The findings from this comparison were thought to form the first constructive step toward the development of an empirically based decision framework for health facility evacuation. About 10 years later, Cullen (2011) sought to examine the strategic emergency preparedness planning changes made by hospitals along the Gulf of Mexico in 5 years after Hurricanes Katrina and Rita. As part of the research, administrators from nine regional hospitals were interviewed to learn about their experiences during Hurricanes Katrina, Rita, Gustav, and Ike. Among the key questions were those related to the emergency strategies that the hospitals have implemented since these events and what has yet to be done. The study also included an evaluation of each hospital's emergency policies and plans, practices and implementation as well as improvements, evacuation versus shelter-in-place strategies, training and drills, supplies, reimbursement, communication and human resource issues. The findings of this assessment were used to identify trends and best practices being used by coastal healthcare facilities, to determine which of these have been put into practice, and where opportunities for future improvements remain.

Among the few formal research efforts to develop systems to model the evacuation processes of hospitals was Taaffe et al.'s (2006) study. In this work, the researchers developed an assessment tool to evaluate the effectiveness of an evacuation plan under various scenarios and resources, including the number of different patient types, the availability of nurses, and the number of vans and ambulances to transport patients. These various scenarios were measured by the average time necessary to carry out an evacuation among these variables and tests.

7.2. Gasoline and other supplies

The excessive delays and congestion associated with the Houston regional evacuation for Hurricane Rita in 2005 revealed numerous problems, many of which were difficult to imagine let alone plan for. Among them was the need to provide in-route services to evacuees, including the delivery and distribution of fuel, water, food, and wrecker services during the evacuation. Based on the Rita event and evacuations since, these types of services are now regarded as more than a mere convenience. They can affect the lives and health of evacuees trapped on congested roads and can serve to clear travel lanes blocked by broken down and fuel-less vehicles to keep traffic moving. The 2009 NCHRP Synthesis on the role of transportation in evacuations (Wolshon, 2009) included a set of survey questions to determine which locations were planning to provide various in-route services, what they were, and how they would be delivered.

The initial reaction to the survey results was surprise because more than half of the survey respondents indicated they were planning one or more of the afore-named services. Overall, 25 of the 36 agencies (about 70%) indicated they had plans in place to provide in-route fuel to out-of-fuel vehicles and 21 indicated that their agencies planned to provide wrecker services to remove disabled vehicles from evacuation routes. From follow up interviews with some of the respondents, vehicle moving would be limited to removing it from the travel lanes to a location where it would not create a blockage. Most often, this would be a shoulder and not a local service facility. The survey questions also permitted a review of the types of agencies who would be leading the coordination of these services. Emergency management agencies were found to most often take the lead in these activities, particularly at the state, rather than local, level. The relatively low numbers in the local agency category were not surprising since several of them indicated they did not have jurisdiction of any roads under their municipality and their portion of the overall evacuation network was quite limited.

Regional scale traffic modeling has also played a role in determining the potential fuel demand during a major regional evacuation. This data was then used to predict potential concentrations of fuel run-outs, and position mobile fuel supply trucks near these locations. Gottumukkala (2012), used surveys of evacuees to determine origins and destinations, numbers of vehicles, routes traveled, and amount of fuel at trip origin. With this information, a fuel demand was computed and was compared to the locations and supplies of fueling stations along evacuation routes in the State of Louisiana to determine the segments of roadway within the state with the highest fuel deficit. Similar efforts to maintain and predict fuel shortages as well as track supplies and evacuees have been developed through academic/commercial partnerships in Texas.

7.3. Schools and children

A 2008 NRC study to assess public knowledge of, confidence in, and decision-making tendencies relative to nuclear power plant protective action orders was conducted by the Sandia National Laboratories (Jones et al., 2008). The research included extensive telephone surveys and focus group sessions of residents living within the ten mile nuclear power plant emergency planning zones (EPZ); focus group interviews with emergency response management and personnel; and cognitive response testing of focus group participants. Included in the research were questions related to the sheltering and movement of school children during emergencies.

Similar to residential facilities, emergency planning procedures require schools within nuclear power plant EPZs to have evacuation plans to transport school children to safe shelter areas. Many parents were also aware of this and indicated they received annual emergency planning information from their children's schools. Despite this planning, the NRC survey showed that many parents will go to schools to pick up children even if told they are already being evacuated and they should not come to the school. A general finding of the research was that parents did not think the schools would safely evacuate their children, often because of a perceived lack of buses. The report further suggested that, in recognition of these opinions and the amount of additional traffic congestion that it would likely create, this is an issue that should be anticipated during nuclear power plant evacuation planning and further validates the ideas behind some of the intermediate trip studies (Liu et al., 2012; Murray-Tuite and Mahmassani, 2003, 2004).

7.4. Elderly and homebound

Issues of evacuating elderly and homebound individuals fall broadly under what has become known as carless evacuation. The concept of evacuating populations without access to personal transportation has been recognized for decades, although the shortcomings of its lack of attention came into sharp focus after Hurricane Katrina. Although it was long realized that substantial portions of the New Orleans population did not have access to private automobiles, relatively little formal planning went into preparing for their movement. While there are many reasons why this was the case, the issue of evacuation is still widely viewed as a personal decision and although more cities and states are providing emergency transportation support it is not always effectively used.

Renne et al. (2011, 2008) have shown that evacuating the elderly and homebound has been and largely remains challenging. Because these individuals are not in formal care center settings, these individuals are often not specifically or individually addressed in any plans. Often, due to issues of maintaining personal privacy, a lack of integration into community activities, or frequent movement, such individuals also do not appear on community assisted evacuation registries. In the past, New Orleans has worked with faith-based and other community groups to develop neighborhood support plans such as "Operation Brother's Keeper" which encouraged neighbors and friends to assist neighbors and friends who may be in need during times of crisis.

The NCHRP Synthesis study (Wolshon, 2009) also surveyed transportation and emergency management agencies on their planning relative to functional needs and carless planning. About 70% of the survey respondents indicated that they are addressing this need. Emphasis was highest at the local level where seven of the eight local-level responding agencies indicated special needs planning. In terms of the specific types of low mobility groups included in the plans, the most commonly cited were frail elderly and persons with disabilities with one or both indicated by about half of the responding agencies. Most of the other categories were also indicated at similar levels. The primary exception was planning for *homeless populations* with only four agencies.

A key part of providing services for non-self-evacuating groups is identifying their numbers and locations. To assess the sizes of these populations another series of questions were posed. The reported percentages of the populations considered to be special needs were considerably higher than was anticipated. Of the 23 agencies responding to the question, six of them estimated their proportion to be in excess of 20% of their total population. Such levels are thought to be significant given the complexity and amount of resources required to evacuate such individuals. Most locations assumed their special needs populations to be between 6% and 10%, although these estimates varied considerably. The data also showed that local and emergency management agencies tended to estimate higher since they "own" the problem and would be expected to estimate more conservatively.

To help in locating special needs individuals, many agencies have developed systems to permit citizens requiring assistance to register for this service. The survey revealed that two-thirds of the responding agencies acknowledged the availability of registration systems in their evacuation plan. This was surprising since such services are typically handled at the local level. The type of systems indicated by these agencies included 211 and 311 telephone systems, internet web pages, public health organizations, and 911 emergency calls. As expected, most agencies estimated the percentage of special needs individuals registering for the service was low, with half of them estimating it to be 10% or less.

7.5. Pets

Following several recent evacuations, it has become increasingly apparent that people evacuate with pets and in some cases livestock, particularly horses. There has also been evidence to suggest that many pet owners are reluctant to evacuate an area if they cannot relocate to a shelter location that permits the housing of pets (Edmonds and Cutter, 2008). As a result, many evacuation plans have been revised to permit the inclusion of animals as part of their operations. To assess the level of provisions for pets in evacuation planning, the NCHRP survey (Wolshon, 2009) investigated how animals are now being considered in plans and how sheltering plans include amenities for pets, livestock, and service companions.

The survey showed that the majority of emergency preparedness agencies now give consideration to pets and service companions at some level and that more than 60% included them in their sheltering plans. It is important to note that all respondents to the survey included accommodations for service animals. Although comparatively fewer in number, some agencies around the country are also beginning to make sheltering provisions for livestock, such as horses, particularly in areas, such as California, where this issue is important to the local populace.

Edmonds and Cutter's (2008) work also resulted in the development of a pet estimation model that can be used to assist local officials in identifying the geographic distribution of families with pets that are likely to require shelter. It was suggested that the use of widely available population datasets and the research methodology would provide the tools necessary to estimate the sheltering needs of pet-owning residents within their jurisdictions.

7.6. Disabled

Efforts to support evacuations for the disabled are another area of ongoing development and improvement. Given the expanding definition of disabled and functional needs populations that may need assistance, it is a significant issue. In fact, a recent survey (Wolshon, 2009), suggested that in many cases, up to 20% of the total population in many areas may be defined as potentially requiring assistance. These figures also do not include other populations such as prisoners, unaccompanied minors (e.g. "latch key" children), and other *protected populations* like halfway house and sheltered residents of group homes and other formal and informal institutional facilities.

Until recently, there were few, if any, formally designated agencies that "owned" this issue. Many of the needs associated with these populations have been identified and discussed in related broader studies of carless populations (e.g. Renne et al., 2011, 2008) and more specifically from a social context (Parr, 1987, 1998). Recent reviews have shown that the evacuation of functional needs and disabled populations will continue to remain a challenge to evacuation planners. This is true not only from the amount of personnel and monetary resources needed to move a small amount of people, but in engaging the persons in need. Several of the recent studies have shown that the majority of special need persons do not register for evacuation assistance (Jones et al., 2008).

7.7. Traffic incidents

Similar to routine periods, traffic incidents should be considered to be a normal and expected part of any evacuation. With the amount of volume and potential for more inexperienced drivers, moving through unfamiliar areas, with unreliable vehicles, the likelihood of in-route incidents might also be expected to be even higher than routine. Given the enormous volumes generated over compressed periods, the effects of on-road incidents also have the potential to be much more wide ranging, long lasting, and safety-threatening. For example, in the Florida Keys where only a single road of egress exists for the majority of an 80,000 person population, a lane blocking incident could endanger the entire evacuation process.

Despite conditions such as these, the effect of traffic incidents are not commonly factored into traffic simulation models or accounted for in evacuation transportation planning. Often this is because these events are deemed as random and there are few reliable methods to predict their location and/or timing. Since many agencies also have plans to utilize motor assistance patrols and other wrecker services, it is possible that incidents are not regarded as significant threat to the overall evacuation process.

To begin addressing this issue, Robinson et al. (2009) examined and quantified the effects of traffic incidents during large-scale evacuations. They used the Virginia Hurricane Emergency Response Plan evacuation plan and simulation to evaluate traffic incidents in Hampton Roads. Simulations were conducted of evacuation traffic over a 70-h period, during which an average of almost 200 randomly selected accidents and 1400 incidents (including disabled and abandoned vehicles) occurred. The incidents significantly extended the travel time of those involved, although the time required to complete the evacuation of the region increased somewhat marginally if evacuees could be rerouted to other roads within the network.

8. Future needs and directions

Despite the extensive body of work highlighted in the previous sections of this paper, significant work remains. Suggestions and needs for future work are mentioned above, but three additional needs and future directions are outlined here. These are just a sample of where evacuation efforts could go.

8.1. Interdisciplinary efforts

As demonstrated by the current paper, much of the work has been conducted in disciplinary silos. However, the trend is growing to create interdisciplinary work with social scientists and engineers. The social scientists have a better understanding of behavior, behavioral theories, and risk perception that are critical to developing quality models for evacuations, while the engineers generally use and develop mathematical and simulation tools to answer practical questions (e.g., how long will it take to evacuate?, what would the time be if we reverse this lane?). Blending these two fields is a good start to addressing the interdisciplinary needs. However, other fields should also be included, particularly those pertaining to the specifics of the different disaster types, which can lead to greater understanding of the impact area and path predictions and perhaps refine the definition of the at-risk area.

Another field that should be incorporated is medicine. This field could offer insights into the medical needs of portions of the population who have additional challenges in an evacuation, the consequences of not evacuating with different post event conditions, and event specific dangers (e.g. chemical inhalation). Medical events, as well as traffic incidents, could

occur during the evacuation process and pose response challenges and traffic delays. Furthermore, the event itself could injure a significant portion of the population, requiring emergency responders and hospitals to receive the injured. The injuries could affect behavioral responses for the families of the injured. The emergency response trips to the scene and to the receiving hospitals are rarely specifically modeled, although plans usually ensure that roads are open for emergency response.

The people who decide to stay for medical or other reasons will need supplies, including, but not limited to medicine, medical supplies, food, and water. Studies related to evacuation could be tied to humanitarian logistics and disaster relief operations to consider the timely delivery of critical supplies. In areas experiencing multiple disasters, the effects of previous experience with effective/ineffective supply delivery could be investigated for influences on evacuation decisions.

8.2. Use of new technologies

Technology has been advancing at a tremendous rate and will continue to offer new potential areas of study for evacuation research. One example is social media. A few researchers have begun examining the influence of social media on the spread of information pertaining to disasters. How that information manifests in actions is still a young field of study. Other examples from the evacuee perspective include mobile devices and in-vehicle devices with access to maps and real time traffic conditions, at least on major roads. Their use in emergency situations is comparatively understudied.

From the emergency management and transportation agencies' perspectives, new data collection methods, such as blue tooth devices and cellular phones, are becoming more prevalent. These devices have not been explored in the evacuation context but potentially reach roads without loop detectors. Additional real time network condition data could in turn be fed back to the evacuees, potentially affecting routing decisions. The data could also be used for post event validation efforts.

Combinations of real-time information on traffic, infrastructure conditions (e.g., damaged, flooded, open), and event conditions may influence all and any of the evacuation related decisions (go/stay, destination, departure time, mode, route, etc.). The collection and dissemination of this data is a young field requiring new models and technologies. Vehicle-to-vehicle and vehicle-to-infrastructure communication may play a role here as it offers a distributed method to collect and communicate information about traffic conditions. How this information is communicated to people who have not departed remains an issue. The influence of the real-time information and dissemination methods on the decisions is also a new branch for exploration and will likely involve the assessment of how they affect people's risk perceptions and evaluation of protective action alternatives.

Real-time information could also be used by evacuation management officials. The traffic and infrastructure conditions may warrant targeted network management strategies, particularly with respect to real-time event information (and short term projections). Keeping track of cumulative volumes may also identify slow evacuation areas and indicate a need for additional warning communication and/or local sheltering resources.

Recent work by [Gudishala and Wilmot \(2010\)](#) examined how the communication of storm track and intensity information and evacuation orders influenced the departure timing of experienced evacuees in coastal regions of southern Louisiana. The research was conducted using video-based theoretical weather statements and evacuation recommendations to influence test subject behavior relative to evacuation departure decision-making. The stated choices of the test subjects were also compared to actual behavior revealed in similar recent storms for analysis. Later work by [Montz et al. \(2013\)](#) and [Zhang et al. \(2013\)](#) used the models developed from these stimulus–response relationships to examine variable–response evacuation planning and Megaregion evacuation traffic processes.

Technological advances will also aid modelers and officials, in general. These modelers could be those who track and predict weather events and with new technology are able to make better predictions or reduce uncertainty. New technologies may offer more widespread or effective warning systems, helping to reach those who need to evacuate and being clear and reassuring to those who do not. Advances in computing technology will continue to allow faster simulations and allow models to contain higher levels of resolution.

8.3. Model refinements and validation

From the literature, it appears that the bulk of data driven evacuation work focused on hurricanes, probably because of the relative frequency and wide spread impact. The applicability of methods developed for one disaster type for other disasters needs to be carefully investigated. Even within disaster types, transferability of results needs to be continually considered.

Several refinements may be made to the demand considerations, but only three are discussed here. Background traffic is a wide open area of study – few, if any, studies have uncovered what reasonable assumptions of these volumes would be. Shadow evacuation is another aspect that needs further investigation. The impact of shadow evacuation is potentially high but the amount of shadow evacuees is inconsistent; methods to predict these evacuee volumes and distributions for the planning process with confidence remain to be developed. Further, the transit based demand as well as pedestrians for smaller scale events have only recently started to be addressed.

From the traffic simulation perspective, four directions could be addressed. First, driving behavior, including spacing and aggressiveness should be addressed for real events. Some initial work has begun in a simulated environment, but the translation to real disasters remains an open issue. Second, improvisation behavior is not captured in simulation tools; some randomness in behavior should be considered. Third, vehicle type mixes and their interactions should be modeled, particularly if

proportions exceed what has been well studied. Vehicle type based capacity reduction factors will need to be incorporated and may be time dependent as vehicle mixes may vary over the course of the evacuation. Finally, route selection and en-route changes need to be based on behavior data driven studies.

Validation for evacuation models is a difficult but important task. With more complexity, the models will become more difficult to validate but this step cannot be ignored.

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